

16th to 22nd July:

Found relevant papers on federated learning x attack / defense mechanisms from AAAI 2026.

23rd to 29th July:

Read these papers thoroughly:

- ShadeEdit: A Utility-Preserving and Defense-Evasive Knowledge Manipulation Attack in Federated LLMs
- BANT: Byzantine Attack via Trial Function and Trust Scores
- Good Gradients Poison Your Model Too: Evading Defenses in Federated Learning via Boundary-Adaptive Perturbation
- FedSEA-LLaMA: A Secure, Efficient and Adaptive Federated Splitting Framework for Large Language Models
- LSHFed: Robust and Communication-Efficient Federated Learning with Locally-Sensitive Hash Gradient Mapping
- Learning to collaborate: An orchestrated-decentralized framework for peer-to-peer LLM federation.

30th July to 5th August:

Read remaining papers:

- Enhanced privacy leakage from noise-perturbed gradients via gradient-guided conditioned diffusion models.
- DoBlock: Blocking Malicious Association Propagation for Backdoor-Robust Federated Learning Under Domain Skew

Implemented four experiments of LSHFed:

- Hamming to Euclidean correlation across an r sweep on within-model gradients.
- Synthetic perturbations (magnitude scale vs directional deviation) to assess which situations the Hamming distance fails to correlate with the Euclidean distance.
- Winner (aggregator selected as the next round's global model by LSHFed) agreement compared at fixed r between Euclidean and Hamming.
- Computational analysis of the correlation vs time- and space- efficiency tradeoff at different values of r .

6th to 12th August:

Implemented two experiments (01 and 02) of GGSS-R:

- Baseline reconstruction attack using locally trained *single-epoch* 3-layered MLP victim model (checkpoint was not provided), arbitrarily chosen target image, and publicly available diffusion model checkpoint.
- Baseline reconstruction using locally trained *three-epoch* victim model, with target image specifically chosen from class 0.

13th to 19th August:

Implemented five experiments (03, 04, 05, 06, 07) of GGSS-R:

- Unperturbed reconstruction diagnosis,
by examining target-gradient validity (non-zero elements and norm),
second order differentiability (what percentage of gradient reach image pixels),
gradient calculation validation (whether the authors' repo calculates distance correctly),
and layer-wise gradient norm of victim model.
Used the authors' reference diffusion checkpoint.
- Unconditional diffusion diagnosis,
to check whether the diffusion model itself can generate images on the natural manifold.
- GSS (guided sampling) with gradient guidance disabled,
to check whether the GSS path is correctly implemented and can generate images on
the natural manifold when activated.
- Direct gradient inversion attack without diffusion,
to check whether gradients were informative enough for reconstruction.
- Gradient identifiability analyses,
to examine per-element gradient statistics,
average pairwise gradient similarity,
class separation,
gradient subspace dimensionality,
clustering tendencies,
gradient-space vs image-space nearest neighbours,
correlation of gradient distance / cosine similarity with pixel-space image similarity,
layer-wise model gradient norms,
gradient-space vs image-space trajectory of ggss-r reconstructions.

Uploaded the first three experiments of LSHFed to GitHub.

- Made the experiments self-contained.
- They were previously dependent on my Google drive, so I had to remove the dependencies and re-execute them.

20th to 26th August:

Implemented the final two experiments (08, 09) of GGSS-R:

- Comparison of reconstruction quality across four custom victim model architectures,
each with two training states (epoch 0 vs epoch 1),
to assess how reconstruction vulnerability changes based on both architecture and
training state.
- Scheduled gradient-guidance rate that decays linearly over diffusion timesteps,
to lower the susceptibility of reconstruction to noise-infused gradients,
keep the sampled images on the natural manifold for longer,
and improve the final reconstruction quality.

27th August to 2nd September:

Uploaded the fourth experiment of LSHFED and the first experiment of GGSS-R to Github.

- Made the experiments self-contained.
They were previously dependent on my Google drive, so I had to remove the dependencies and re-execute them.

Also made experiments 02, 03 and 09 of GGSS-R self-contained, but did not yet upload them.

3rd September to 7th September:

Uploaded all experiments (from 02 to 09) of GGSS-R to Github.

- Updated their code descriptions and readme files.
- Cleaned up the repo structure.

Experiments 04, 05, 06, 07 and 08 could not be made self-contained as they received a lot of inputs from my personal Drive.